

Opening the City: Trade Liberalization and Urban Expansion in Brazil

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Abstract

I study the long-term consequences of Brazil's trade liberalization in the 1990s on urbanization patterns in tariff-exposed regions. I exploit the timing of the tariff reform and the industrial composition of each local labor market to construct a shift-share measure of tariff exposure at the local labor market level. The main findings reveal that regions harder hit by tariff reductions experienced a substantial increase in their urban footprint relative to less exposed regions. This result is robust to several specifications and controls. My proposed mechanism combines the long-term adverse effects on local labor markets with the formation of slums within the most exposed regions.

Keywords: Trade, Urban, Slums

JEL Codes: F14, F16, R12, R23

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1 Introduction

According to UN (2024), the population of Latin America and the Caribbean is projected to increase from 663 million in 2024 to 717 million by 2040, while the population of low-income countries is expected to grow by approximately 45% over the same period. As documented in UN-Habitat (2025), 1.2 billion people currently live in slums, and 2 out of 5 people worldwide still lack access to safely managed sanitation services. Population growth alone, however, does not explain why people end up in informal settlements. Slums are not a natural byproduct of population growth; they are, at least in part, the endogenous outcome of labor market frictions and the location decisions of firms and workers interacting under imperfect housing markets (Moretti 2010; Krugman 1991; Glaeser, Kolko, and Saiz 2001). Understanding what drives people into informal housing therefore requires understanding what drives them out of formal labor markets.

In this paper, I ask: what happens to the spatial equilibrium of cities when local labor markets are hit by a trade shock? Using, Census Data, high-resolution satellite data from Projeto MapBiomias (2024) and following recent evidence on the negative local labor market consequences of trade liberalization (Dix-Carneiro 2014; Topalova 2010; Kovak 2013; Ponczek and Ulyssea 2022), I leverage variation in industry composition across local labor markets to identify the effect of tariff-induced labor demand shocks on the expansion of the urban footprint across Brazilian microregions. To the best of my knowledge, this is the first paper to combine high-resolution satellite data with a shift-share identification strategy to provide evidences on how the spatial distribution of urban areas responds to trade-induced labor market shocks.

My evidence shows that, on the extensive margin, the urban footprints of cities more exposed to trade liberalization grew faster than those of less exposed cities. Trade-exposed regions experienced simultaneous outward expansion of the urban footprint and growth in slum settlements at the urban fringe. This pattern is consistent with a reduction in household density and an increase in sprawl, driven by the dynamics of city-border growth described in Henderson, Regan, and Venables (2021): building volume delivered by slums increases over time, and new slums appear near the expanding city edge, where land is relatively cheap and informality faces less regulatory pressure. Moreover, my findings using Projeto MapBiomias (2024) suggest that slum formation in Brazilian regions is partly driven by a combination of permanently worsening local labor market conditions, weak land regulation, and housing pressure. This result is supported by the event study estimates, which display no pre-existing trends prior to the trade liberalization, strongly suggesting a causal interpretation. The pattern of slum formation is further consistent with weak land regulation in trade-exposed regions: land with slopes above 30% — nominally subject to building restrictions — is occupied under housing pressure and converted into informal settlements. Evidence from the 1991 and 2000 Brazilian Household Censuses confirms that slums grew significantly larger in regions more exposed to the tariff reform relative to less exposed regions over the same decade. These results are robust to the inclusion of controls from both the 1991 and 1980 Brazilian Censuses and to a variety of functional forms for the dependent variable.

My findings contribute to two main strands of the urban and trade literature. First, I contribute to the extensive literature on the local labor market effects of trade-induced shocks (Dix-Carneiro 2014; Dix-Carneiro and Kovak 2017; Kovak 2013; Topalova 2010; Autor, Dorn, and Hanson 2021; Ponczek and Ulyssea 2022; Felix 2021; Fan 2019). Using the same shift-share identification strategy as Dix-Carneiro and Kovak (2017), I show that the permanent deterioration of local labor market conditions induced by the tariff reform generated a measurable spatial response: the expansion of informal settlements and urban footprints in more exposed regions. My novel contribution is to document that slum formation is a spatial equilibrium object of trade-induced labor market shocks, a margin of adjustment that has been largely overlooked in the existing literature and recently studied by Mariano and L. Pereira (2025). Second, I contribute to the growing literature on urbanization responses to trade shocks (Mariano and L. Pereira 2025; Nagy 2022; Nagy 2023; Bonadio et al. 2023). My novel contribution to this strand is to use high-resolution satellite data on urban land cover — measured at 30-meter resolution across all Brazilian municipalities — to document how the urban footprint evolves in response to local labor market shocks induced by trade liberalization. To the best of my knowledge, this is the first paper to bring this type of spatially granular land-use data to bear on the question of how trade-induced shocks reshape the physical geography of cities in a developing country context.

The rest of this paper is organized as follows. Section 2 provides historical background on Brazil’s 1990s trade liberalization, describing the structure of the tariff reform and its labor market consequences. Section 3 describes the data sources, including the satellite-based urban footprint data from MapBiomas, the IBGE household censuses, and the tariff measures. Section 4 presents the shift-share measure of regional tariff exposure and the empirical strategy. Section 5 presents the main results on urban footprint expansion. Section 6 examines the mechanisms, focusing on slum formation and land regulation. Section 7 concludes.

2 The 1990s Trade Liberalization in Brazil

Before the 1990s, the Brazilian economy was heavily protected by a series of tariffs, especially the ad valorem tariffs introduced by the Brazilian government in 1957 (Hay 2001). At that time, most of the industrial sectors in Brazil operated under a redundant system of tariff protection. Brazilian firms, particularly those in the consumer goods sector (especially durables), faced a high level of trade barriers. This protectionist environment at the time was associated with low capital-to-output ratios, low productivity, low wages, and limited economies of scale (Pinheiro and Almeida 1994).

This protectionist scenario began to shift in 1988, during President Sarney’s administration, which outlined a new industrial policy aimed at reforming trade protection. The main goal was to reduce redundancy in the tariff structure. Despite the comprehensive nature of this policy, few practical reforms were actually implemented. As pointed out by Kume, Piani, and Souza (1990), this initial movement toward lower tariff barriers was insufficient to significantly reduce effective protection levels across industries, mainly due to strong opposition from

industrial lobbies.

In the 1990s, President Fernando Collor revived the reform agenda, this time with enough political strength to ensure a substantial reduction in overall industrial protection. The reform targeted a broad list of 1,300 previously prohibited products. The new tariff reform program was designed to be implemented in stages, starting in February 1991 and the following in January 1992, October 1992, and July 1993. As noted by Kume (1989) and Hay (2001), all tariffs on machinery and equipment not produced in Brazil were immediately reduced to nearly zero. In summary, the goal was to bring all industrial tariffs down to an upper limit of 20%.

A notable set of facts relevant to our context is that the 1990s brought economic stress beyond the negative industrial labor market effects of the trade reform, compounded by a severe recession. The financial crisis of 1998 further contributed to a period of economic instability in the Brazilian labor market (Neri, Camargo, and Reis 2000). A countervailing force during this turbulent period was the expansion of informality and the service sector in Brazil (Ponczek and Ulyssea 2022). This is particularly relevant in our urban context: according to the 1991 Brazilian Census, approximately 30% of the employed population living in slums worked in the service sector, 19% in industry, 15% in commerce, and 21% were informally employed.

3 Data

Tariffs. For the nominal tariff data, we rely on Kume, Piani, and Souza (1990), which provides sector-level tariff rates for the period 1987–1998. These tariffs are averaged across sectors using the input-output matrix from the Brazilian Institute of Geography and Statistics (IBGE). We use this data to construct our trade liberalization measure across industries and regions, following Dix-Carneiro and Kovak (2017) and Kovak (2013).

Urbanization. We use spatial data at 30-meter resolution from the MapBiomass platform (Projeto MapBiomass 2024), Collection 10.¹ The methodology underlying the urban footprint maps is described in Souza Jr et al. (2020), who use Landsat satellite imagery and machine learning techniques within the Google Earth Engine platform to reconstruct annual land use and land cover classifications at 30-meter spatial resolution since 1985. From this database, we use the category *Non-Vegetated Urban Area*² as our main measure of urban area. This category captures the bulk of urban infrastructure, including buildings, roads, industrial facilities, and other constructed surfaces. The MapBiomass classification scheme is broadly consistent with those of the Food and Agriculture Organization (FAO) and IBGE (Souza Jr et al. 2020). The same data have been used in recent work to measure urbanization changes following land-slides (Lima et al. 2025), to study agricultural conflicts in rural Brazil (Falcone and Rosenberg 2025), and to estimate the equilibrium effects of road networks on deforestation (Gollin and

¹<https://brasil.mapbiomas.org/wp-content/uploads/sites/4/2025/08/ATBD-Collection-10-v2.pdf>

²The Non-Vegetated Urban Area category in Souza Jr et al. (2020) also includes areas such as beaches and dunes, rocky outcrops, and mining sites. A detailed description of the structural procedure for the urban and other MapBiomass modules is available at <https://brasil.mapbiomas.org/download-dos-atbds-com-metodo-detalhado/>.

Wolfersberger 2025).

Figure 1 and Figure A.1 illustrates how our urban footprint measure works, using the city of Volta Redonda in the state of Rio de Janeiro as an example. Volta Redonda was among the municipalities most exposed to the tariff reform according to our exposure measure described above. The figure displays the urban footprint in 1991 and in 2000, a decade after the onset of the reform. Blue areas represent the urban footprint as of 1991, while red areas capture the expansion between 1991 and 2000. The pattern of red areas concentrated at the city's periphery is suggestive of outward urban expansion along the city's borders. This is not causal evidence, but it illustrates how the urban dynamics we study are reflected in our outcome variable. Another relevant consideration regarding the Projeto MapBiomass (2024) data is that urban area classifications are constructed using the 2022 Brazilian Census tract boundaries defined by IBGE. These boundaries are held fixed throughout the entire panel period from 1985 to 2023. This does not concern us, as the fixed geographic container allows us to track the evolution of urban land cover within each census tract from the first year of the panel onward.

1980, 1991, and 2000 IBGE Household Censuses. We use the IBGE household censuses to construct a measure of household structural composition following IBGE (2024), and to assess the share of households classified as slums within each urban census tract. Using the 1991 and 2000 censuses, we estimate long-difference specifications for each outcome, with the 1991 baseline value of the dependent variable included as a control. We additionally use the 1980 Census to construct pre-reform controls in our main specifications.

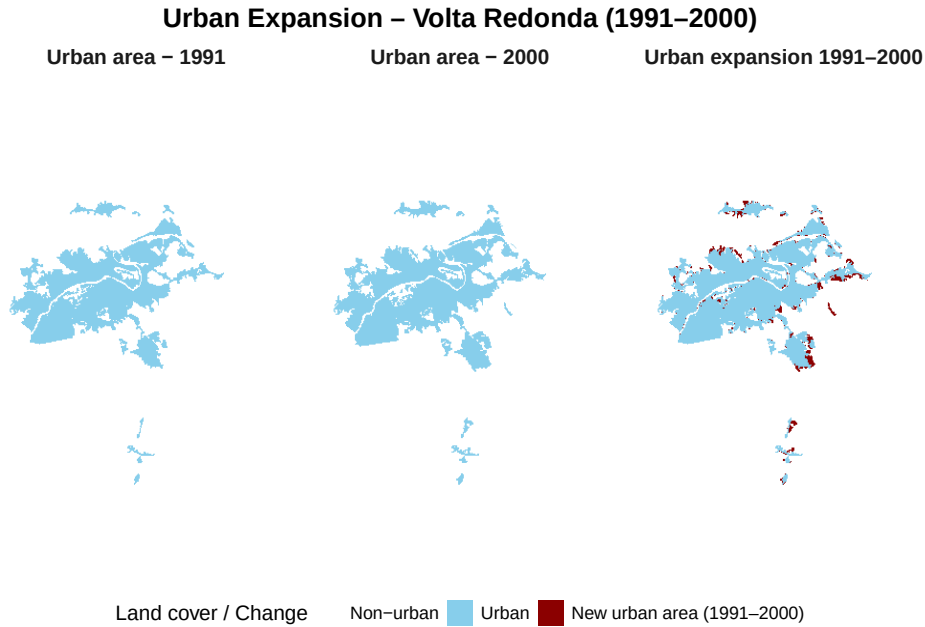


Figure 1: Long difference 2000 - 1991 of the urban footprint for the city of Volta Redonda

Note: The panels show the urban footprint of Volta Redonda - RJ in 1991. The rightmost panel displays the long difference between the two years, highlighting areas that became urbanized over the period.

4 Measuring Municipal Exposure to Trade Liberalization

We construct a measure of exposure to tariff reductions at the microregion level³ based on the approaches of Dix-Carneiro and Kovak (2017) and Kovak (2013). We use 486 microregions, and throughout the paper we use the terms microregion and region interchangeably. Since our spatial data from Projeto MapBiomass (2024) is organized at the municipal level, we work with the 5,565 municipalities recorded in the 2010 Brazilian Census, but aggregating at the level of our exposure measure within each regression.⁴ We apply the same restriction to account for the creation and reorganization of regions over time: between 1980 and 2010, the number of municipalities increased by approximately 40 percent. When aggregating to the microregion level, we consider the total urban footprint of all municipalities within each microregion.

We follow Kovak (2013) to measure the impact of the tariff shock on labor demand across industries within each microregion. The underlying idea is that regions with a higher initial concentration of employment in industries facing larger tariff cuts will be more exposed to the reform, and will consequently experience a larger negative shock to local labor demand. Following Kovak (2013), we have:

$$\sum_i \beta_{mi} \hat{P}_i \quad \text{where} \quad \beta \equiv \frac{\lambda_{mi} \frac{1}{\varphi_i}}{\sum_j \lambda_{mj} \frac{1}{\varphi_j}} \quad (1)$$

where $\hat{P}_i$ represents the liberalization-induced price change faced by industry i , and Equation 1 represents a weighted average of these price changes across tradable industries.⁵

Following Dix-Carneiro and Kovak (2017), we construct the tariff shock measure in Equation 2 by defining the regional tariff reduction (RTR_m), which captures the tariff change relative to 1990 for each tradable industry in each region:

$$RTR_m = \sum_{i \in \mathcal{I}} \beta_{mi} \times (Tariff_{1995,i} - Tariff_{1990,i}) \quad (2)$$

In Equation 2, m indexes microregions, i indexes industries, and β_{mi} is the baseline employment share of industry i in region m . We set the empirical baseline to the 1980s, using pre-reform census data to capture the industrial composition of each municipality prior to the tariff shock. By construction, RTR_m is more positive in regions facing larger tariff reductions, which simplifies the interpretation of our results since nearly all regions experienced tariff declines during the liberalization reform.

³Our primary measure of tariff change is at the microregion level, following the IBGE classification IBGE – Instituto Brasileiro de Geografia e Estatística (1990), which defines 558 microregions.

⁴Over the period of our study, the spatial boundaries of Brazilian municipalities changed substantially. We therefore use the same harmonization procedure as Dix-Carneiro and Kovak (2017) that maintains consistent geographic units between 1985 and 2010, accounting for border changes within each microregion. This harmonized unit is referred to as an MMC (minimum comparable microregion).

⁵We follow Kovak (2013) in placing greater weight on industries that account for larger shares of initial regional employment. Here, m indexes regions and λ_{mi} is the share of regional labor initially allocated to tradable industry i . We focus exclusively on the tradable sector, dropping non-tradable industries under the assumption that non-tradable prices move proportionally with tradable prices.

4.1 Empirical Strategy

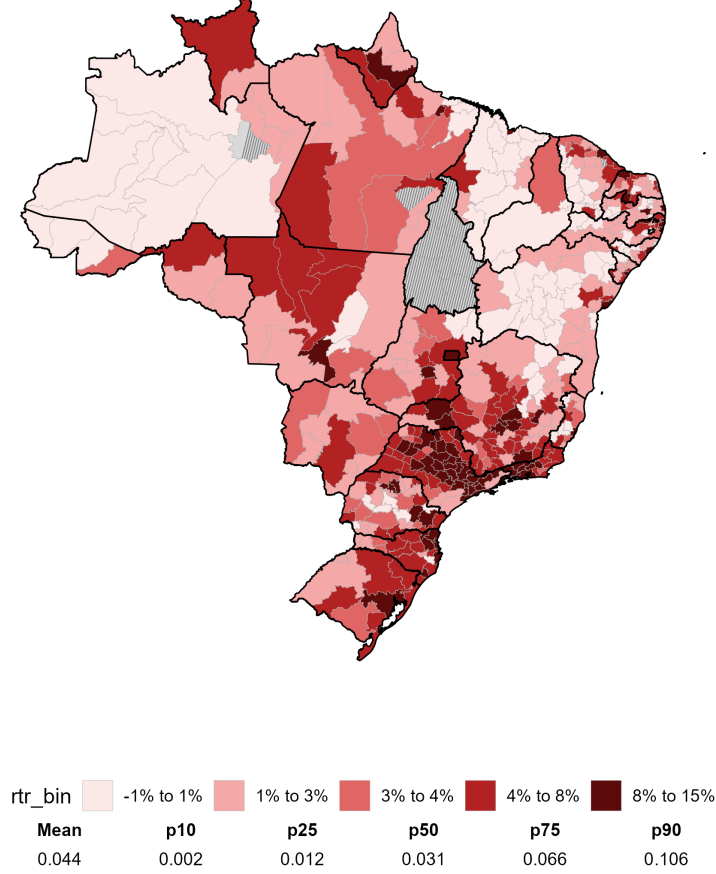


Figure 2: Regional Tariff Reductions

Notes: Local labor markets reflect microregions (MMCs) defined by IBGE, aggregated slightly to account for border changes between 1985 and 2010. Regions are colored based on the regional tariff reduction measure, RTR_r . Regions facing larger tariff cuts are shown in darker red, while regions facing smaller cuts are shown in lighter shades. Black lines represent state borders, gray lines represent microregion borders, and cross-hatched microregions are omitted from the analysis. These microregions were either (i) part of a Free Trade Area, (ii) part of the state of Tocantins and not consistently identifiable over time, or (iii) not included in the RAIS sample before 1990 (Dix-Carneiro and Kovak 2017).

Using the measure of trade exposure defined in Equation 2, we assess how the urban footprint within each region changed following exposure to trade liberalization. Our first specification, shown in Equation 3, follows the approach of Dix-Carneiro and Kovak (2017). It includes MMC and year fixed effects, along with a pre-reform outcome trend for the period 1985–1990 to control for pre-existing urbanization patterns in each region⁶. A post-reform indicator aggregates all periods after the first year of the reform, 1991, allowing us to estimate the overall long-run consequences of the reform for more exposed regions. Our second specification, presented in Equation 4, estimates the dynamic yearly effects of the liberalization measure

⁶In each regression we need to interact the outcome pre-trend with year dummies to avoid multicollinearity with the mmc fixed effects.

RTR on the urban footprint of each region, using the same set of controls as in Equation 3.

In both specifications, the coefficient θ has an interpretation very similar to that in a standard continuous difference-in-differences design: it captures the marginal response of the outcome to a one-unit increase in RTR . While the two equations share a similar conceptual foundation, they differ in their temporal aggregation. The coefficient in Equation 3 is equivalent to a standard two-way fixed effects estimator, pooling all post-reform periods into a single post indicator across the entire support of RTR . Equation 4 recovers the same θ on a year-by-year specification (Callaway, Goodman-Bacon, and Sant’Anna 2024). It is important to note that our estimates do not identify the *effect* of increasing liberalization in absolute terms; rather, they capture the relative response of the urban footprint to marginal differences in tariff exposure across regions. In the results section, we present estimates both in terms of marginal increases in the exposure measure and in terms of movements along the empirical distribution of RTR .

$$Y_{mt} = \alpha_m + \gamma_t + \theta \cdot RTR_m \cdot Post_{1991} + \varphi_t (Y_{m,1990} - Y_{m,1985}) + \varepsilon_{mt} \quad (3)$$

$$Y_{mt} = \alpha_m + \gamma_t + \sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \theta_k (RTR_m \times \mathbf{1}\{t - 1991 = k\}) + \varphi_t (Y_{m,1990} - Y_{m,1985}) + \varepsilon_{mt} \quad (4)$$

In both specifications, the main outcome variable is the urban footprint of each region, measured as the structurally urbanized area within urban census tracts from Projeto Map-Biomass (2024). Specifically, Y_{mt} captures the built-up area (including buildings, roads, and facilities) within the official urban boundaries of each municipality, aggregated to the microregion level. Our preferred specification uses the outcome in levels, measured in hectares. Standard errors are clustered at the MMC level throughout, to account for spatial correlation and for labor mobility across local labor markets (Abadie et al. 2023; Ferman 2019).

A important identifying assumption of our empirical design is that the tariff reform was not driven by the spatial equilibrium of each region. That is, pre-existing trends in urban footprint growth within each region should not predict the level of tariff exposure RTR_m . For this assumption to be violated, there would need to exist a spatial determinant that simultaneously shaped both local urbanization dynamics and the industry-level tariff changes that each region faced, even though the reform was set at the national level. Urban equilibria are highly endogenous to firm and household decisions in labor and housing markets: residential location decisions depend on current and expected income, commuting costs, and local amenities (Mills 1967), so any confounder would need to have been triggered precisely at the time of the reform and to have differentially affected the industries concentrated in each region. In operational terms, this assumption requires the absence of pre-existing trends in the event study estimates from Equation 4: all coefficients for periods prior to 1991 should be statistically indistinguishable from zero.

5 Main Results

We estimate Equations 3 and 4 under multiple specifications, reporting the results in Table 1 and Figure 3. We begin the discussion of our results with the event study estimates for urban area expansion in urban census tracts, shown in Figure 3. In our preferred specification, the estimates display a clear and consistent pattern: regions facing larger tariff exposure underwent a structural shift in their urbanization dynamics in terms of urban footprint expansion. This is supported by the sharp increase in urban expansion rates precisely at the time of the reform and by the absence of any pre-trends in the pre-tariff reform period. The dependent variable in Figure 3 is the urban footprint area in hectares for each MMC. These results are robust to several alternative specifications, including the log transformation of the dependent variable, normalization by the 1991 urban area, and the inclusion of Census controls.

Table 1: Effect of RTR on Urban Patterns

	Urban (ha)	Urban (log)	$\frac{\text{Urban}}{\text{Area}_{1991}}$	Δ Density (log)	Sprawl
RTR_m	1375.093** (562.498)	2.344*** (0.808)	663.045*** (140.117)	-0.562*** (0.162)	0.757* (0.307)
MMC	382	382	387	485	485
R^2	0.719	0.437	0.732	0.161	0.129
Fixed Effects & Controls:					
Unit (MMC)	Yes	Yes	Yes	No	No
Year	Yes	Yes	Yes	No	No
Cluster	MMC	MMC	MMC	MMC	MMC
Census controls	Yes	Yes	No	No	No
Baseline control	No	No	No	Yes	Yes

Note: This table reports estimates of the effect of tariff reform on urban expansion patterns. Columns (1)–(3) report panel estimates of Equation 3 with MMC and year fixed effects. Column (1) uses urban area in hectares with 1980 Census controls. Column (2) uses the log of urban area. Column (3) uses urban area as a share of total MMC area in km² from 1991. Columns (4) and (5) report cross-section OLS long-difference estimates from 1991–2000. Column (4) is the log change in urban density (urban households per urban hectare). Column (5) is the sprawl index (urban area growth rate minus urban household growth rate). Both columns (4) and (5) include in its specifications outcome-specific baseline controls. Standard errors are clustered at the MMC level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Turning to the point estimates in columns (1) and (2) of Table 1, the magnitude of the long-run impact of tariff reductions on urban expansion is large. A one-unit increase in our tariff exposure measure is associated with an accumulated increase of 1,375 hectares in the years following the reform relative to regions facing smaller tariff reductions. In terms of the empirical distribution, a municipality moving from the median to the 95th percentile of tariff exposure would accumulate approximately 125.8 additional hectares of urban area within its urban census tracts. One striking fact is that this effect is roughly equivalent to the entire physical footprint of Rocinha, the most populous favela in Brazil, home to over 72,000 residents in Rio de Janeiro. Our results should be interpreted with caution, as we cannot disentangle the composition of urban footprints or identify the intensive margin of urban expansion. Since we only observe

the urban footprint of each region, our estimates do not provide direct evidence on population density within city boundaries or on changes in land use intensity in already-urbanized areas.

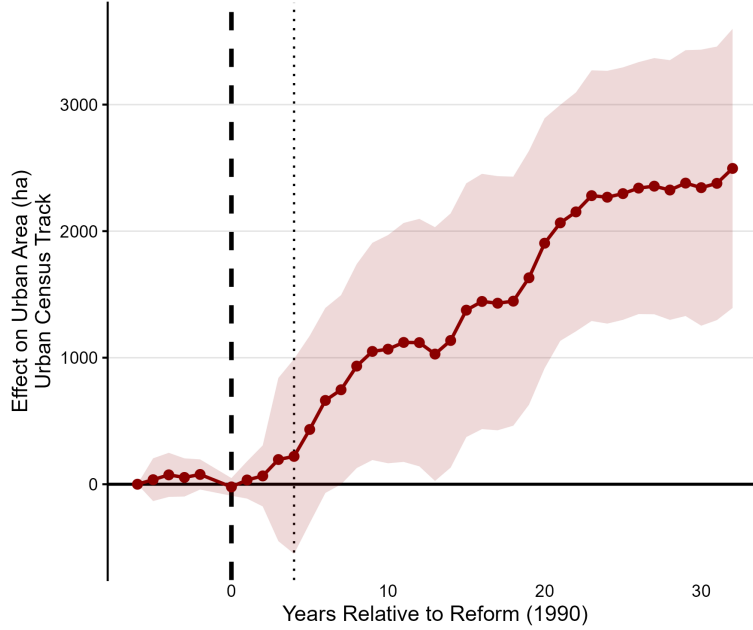


Figure 3: Long Term Effect of the Tariff Reform on Urban Patterns at Micro-region Level

Note: The figure shows the estimation of equation 4 at the micro-region level, with the dependent variable as the urbanized area in hectares within the urban boundaries of each municipality. All regressions include micro-region and year fixed effects along with the 1985–1990 pre-trend of the dependent variable, and 1980 Census controls (per capita income, urban population share, white population share, and higher education share). The treatment variable is at the micro-region level. The baseline period is set to 1990.

Our results in Table 1, columns (4) and (5), provide evidence on the mechanism through which tariff exposure shaped urban expansion. Column (4) uses the 1991 and 2000 Brazilian Household Censuses together with urban footprint data from Projeto MapBiomass (2024) to estimate changes in household density within urban census tracts. The dependent variable is the log change in the ratio of urban households to urban area (in hectares), so a negative coefficient indicates that urban area expanded faster than the urban household stock in more exposed regions: consistent with outward expansion rather than densification. Column (5) complements this evidence with a sprawl index, defined as the difference between the growth rate of the urban footprint and the growth rate of urban households. A positive coefficient indicates that urban area expanded disproportionately relative to population growth, a pattern consistent with urban sprawl (Mills 1967). We rationalize these findings through the following mechanism: the tariff reform induced a negative shock to formal employment in exposed regions (Ponczek and Ulyssea 2022), increasing the demand for lower-cost housing at the urban fringe, where land regulation tends to be weaker and informal construction more accessible (Henderson, Regan, and Venables 2021). City borders thus become the margin of adjustment, absorbing displaced workers through informal housing rather than densifying existing urban cores. We acknowledge, however, that we cannot directly test whether the new urban area was formed contiguously at the city border or through leapfrog development further out. It is also worth noting that our density measure captures the stock of households rather than population. That

is, we estimate how the number of dwelling units per unit of urban area changed, not the number of residents. Therefore one should interpret the estimates in columns (4) and (5) as suggestive of outward urban expansion, noting that the coefficients become less precise when 1991 Census controls are included. We acknowledge that, notwithstanding the high spatial resolution of the data from Projeto MapBiomass (2024), it does not allow us to precisely identify the spatial composition of urban footprint growth within each region. Nevertheless, we provide additional evidence in favor of a border-expansion mechanism. First, we estimate Equation 4 using the log of agricultural area as the dependent variable and find no evidence that regions more exposed to the tariff reform experienced long-run growth in agricultural land, as shown in Figure A.3. This rules out the possibility that the expansion of the urban footprint came at the expense of agricultural land being reclassified as urban, and is consistent with dispersion forces operating within cities rather than urban-agricultural boundary shifts⁷. Second, one might argue that intra-urban green areas were encroached upon due to housing pressure in more exposed regions. To address this, we also estimate Equation 4 using intra-urban green area as the dependent variable and find no evidence that cities in more exposed regions experienced a reduction in within-city green space, as shown in Figure A.2. Taken together, our evidence suggest that neither agricultural expansion nor the occupation of intra-urban green areas can account for the urban footprint growth documented in our main estimates, further supporting a mechanism of outward expansion at the city border.

6 Mechanisms: Slums and Land Regulation

According to the 1991 Brazilian Census, approximately 5% (4.5 million) of the Brazilian population lived in subnormal agglomerates⁸, commonly referred to as slums. In 1991, Brazil had 3,187 of this type of settlements; by 2000, this number had risen to 3,906; and by 2010, the Demographic Census recorded 6,329 agglomerates (Pasternak and D’Ottaviano 2016). Moreover, between 1991 and 2000, the total number of households grew at an annual rate of 0.88%, while the number of households in slums grew at 4.18% per year, nearly five times faster. As suggested by Ferreira, Monge-Naranjo, and L. T. d. M. Pereira (2025), the combination of high housing costs in the formal parts of cities with the pecuniary and non-pecuniary costs of living in slums constitutes one of the main determinants of the sorting of population across neighborhoods, driving households toward informal settlements.

Motivated by this evidence, we use the 1991 Brazilian Household Census to examine how the effect of the tariff reform varies across regions according to the distribution of rental values. With the 1991 Brazilian Census, we can observe the rent paid by each household and split the sample into quartiles, running the same specification as in our baseline estimates. The results

⁷Note that our analysis is restricted to urban census tracts. While we find evidence that rural census tracts experienced urbanization in more exposed regions, we do not find the same pattern within urban census tracts.

⁸A household is considered to be located in a subnormal agglomerate if the census tract in which it is situated lacks essential public services (water supply, electricity, trash collection, and sanitation), occupies or has recently occupied (within the past ten years) land without formal property rights, and constitutes a dense and disorganized settlement.

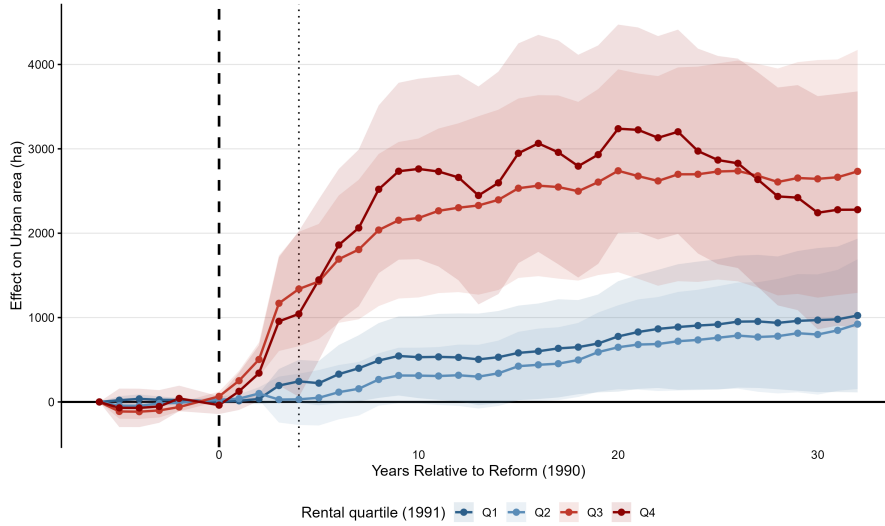


Figure 4: Long Term Effect of the Tariff Reform on Urban Patterns by Rent Quartile from 1991 Census

Note: The figure shows the estimation of equation 4 at the micro-region level, with the dependent variable as the urbanized area in hectares, with the sample split by rental share quartile from the 1991 Brazilian Census. All regressions include micro-region and year fixed effects along with the 1985–1990 pre-trend of each dependent variable. The treatment variable is at micro-region level. The baseline period is set to 1990.

are displayed in Figure 4. As shown, the magnitude of the tariff reform effect on urban area expansion is systematically larger in regions with higher initial housing prices. This effect is persistent and nearly twice as large for regions in the top quartile relative to those below the median. We use this heterogeneity to motivate our main mechanism: slum formation.

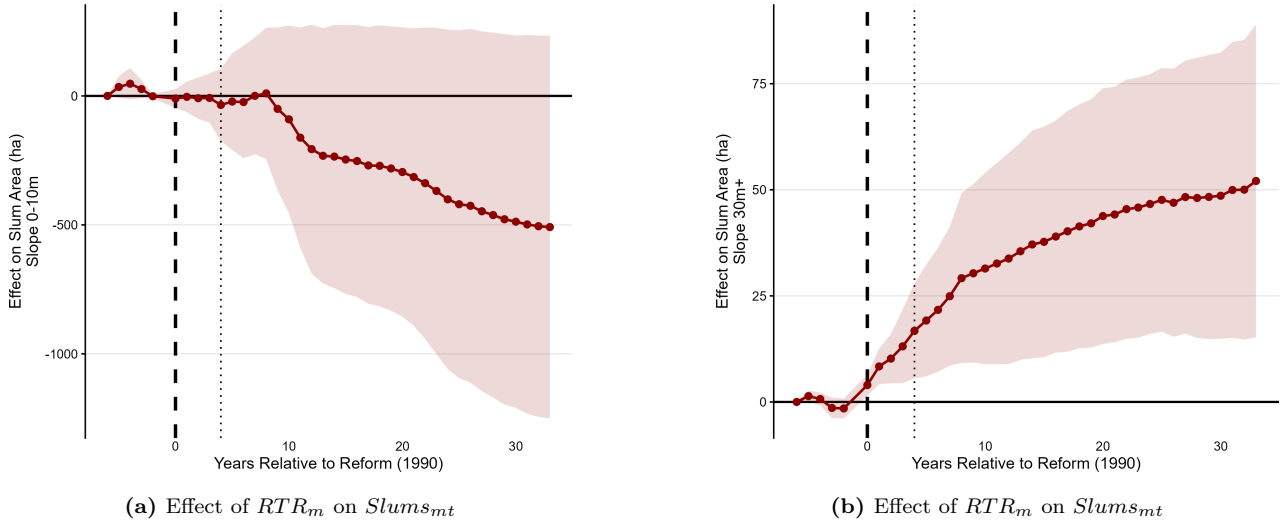


Figure 5: Long Term Effect of the Tariff Reform on Slums Formation

Note: The figure presents the estimation of equation 4 at the micro-region level, with the dependent variable measured in levels. We include year and municipality fixed effects, as well as interactions between pre-trends and year dummies. All estimations are using 1980 Census controls (per capita income, urban population share, white population share, and higher education share) interacted with a post-reform indicator. In Figure 5a, the dependent variable is the structural urban area of slums located in areas with slope up to 10%. In Figure 5b, the dependent variable is the structural urban area of slums located in areas with slope above 30%.

As we discussed in previous sections, the reduction in import tariffs led to a contraction in local labor demand and wages in exposed regions. The evidence in Dix-Carneiro (2014) and

Dix-Carneiro and Kovak (2015) further suggests that these effects were long-lasting in harder-hit regions, compounded by structural frictions to labor mobility across areas. We therefore argue that part of the faster urban expansion observed in tariff-exposed regions -mainly in slums- is driven by two forces: weak land use regulations and housing supply pressure on one hand, and high rental prices combined with the worst labour market conditions on the other. In Brazil, a mandatory regulation⁹ restricts urban construction on terrain with a slope equal to or above 30%. This restriction reflects Brazil’s long history of natural disasters, particularly in poorer urban settlements such as favelas (P. J. Alves, Andrade Lima, and Emanuel 2022). Nevertheless, despite being formally prohibited, these steep areas remain open to informal occupation and serve as a primary locus of slum formation in Brazilian cities. Furthermore, in Brazil every municipality with more than 20,000 inhabitants is required to maintain an urban master plan, revised every ten years. This means that despite the existence of a federal norm restricting construction in risk areas, individual regions retain discretion to adjust their land use policies, potentially easing access to and occupation of high-risk terrain. This requirement was introduced by the City Statute (*Estatuto da Cidade*) in 2001, which mandated that all eligible regions adopt their own urban master plan by 2006. However, since full enactment only took effect in 2006, and notwithstanding the positive evidence of the policy (G. Alves 2025), a thorough investigation of the consequences of these local planning policies falls beyond the scope of the present study.

Table 2: OLS Estimation: Effect of RTR Post-Reform on Slum Area by Terrain Slope

Dep. Variable:	Up to 10m	Up to 10m (ctrl)	Above 30m	Above 30m (ctrl)
<i>RTR_m</i>	53.097 (71.758)	−263.419 (279.411)	22.678*** (6.151)	35.865* (15.032)
Observations	22,960	25,560	22,960	25,560
MMC	247	240	247	240
<i>R</i> ²	0.851	0.890	0.831	0.821
Fixed Effects & Controls:				
Unit (MMC)	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Cluster	MMC	MMC	MMC	MMC
Census controls	No	Yes	No	Yes

Note: This table reports OLS estimations of the effect of the tariff reform on slum area by terrain slope category in hectares. Columns (1) and (3) report the baseline specification without controls. Columns (2) and (4) add 1980 Census controls (per capita income, urban population share, white population share, and higher education share) interacted with a post-reform indicator. All specifications include MMC and year fixed effects as well as slope-specific pre-trend controls interacted with year fixed effects. Standard errors are clustered at the MMC level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 5 presents our estimates of the long-run consequences of the tariff reform on slum formation, separately for areas with terrain slope below 10% and above 30%. The event studies show that regions facing larger tariff reductions experience faster growth in slum areas, with

⁹Law No. 6.766, of December 19, 1979. Available at: https://www.planalto.gov.br/ccivil_03/leis/16766.htm

the effect being particularly pronounced in high-risk terrain. Table 2 presents the main point estimates of the change in slum formation patterns in tariff-exposed regions. Our estimates suggest that tariff-exposed regions experienced faster slum expansion, notably in risk areas, which is consistent with the fact that unoccupied land within cities is formally prohibited from being legally settled. The coefficient is large and statistically significant, suggesting that a one unit increase in tariff reduction is associated with an expansion of 22.7 hectares of slum area in steeper terrain. In contrast, the estimates for slum areas with slopes below 10% are not statistically significant. This finding is consistent with the notion that flatter areas are occupied first due to lower construction and occupation costs, so we would not expect to find significant effects on slum expansion in these zones. Overall, these findings are consistent with our reasoning that the two slope categories face different occupation pressures in tariff-exposed regions, where more vulnerable households are pushed toward the steepest and most precarious areas of each municipality, where land remains informally accessible despite federal land use regulations.

Table 3: Pooled Estimates: Effect of RTR on Urbanization Patterns on 2000–1991 Household Censuses

	Slum HH	Precarious HH
$RTR_m \times \mathbf{1}[2000]$	0.296** (0.135)	0.000 (0.029)
Observations	776	776
MMC	388	388
R^2	0.825	0.725
Mean (1991)	0.015	0.015
Fixed Effects & Controls:		
Unit (MMC)	Yes	Yes
Year	Yes	Yes
Cluster	MMC	MMC
1991 Baseline control	Yes	Yes
Census controls	Yes	Yes

Note: This table reports panel estimates of Equation 3 for outcomes from the 1991 and 2000 Brazilian Household Census, restricted to urban census tracts. The dependent variable is the number of urban households in precarious conditions and in slums normalized by the total urban households in 1991. Precarious households are defined as those lacking all 5 basic services simultaneously: piped water, adequate sanitation, waste collection, electricity, and household density below 3 residents per bedroom. Census controls include per capita income, urban population share, white population share, and higher education share from the 1991 Census, all interacted with the post-2000 indicator. Mean (1991) reports the unweighted average of the dependent variable in the baseline year. Standard errors are clustered at the MMC level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Moreover, we pool the 1991 and 2000 Brazilian household censuses to examine microeconomic evidence on housing quality and informal settlement growth. We construct two urbanization outcomes at the micro-region level. The first captures the intensive margin of precariousness: the number of households classified as precarious, normalized by the 1991 total number of households.¹⁰ The second captures the extensive margin: the number of households located

¹⁰A household is classified as precarious if it simultaneously lacks all five of the following basic services: piped

in subnormal agglomerates, also normalized by the 1991 household total, following the IBGE classification of census tract type (IBGE 2024). Both variables are estimated using Equation 3 in a two-period panel with MMC and year fixed effects and control of 1991 baseline stock of each variable interacted with the post variable.

We present our estimates in Table 3, which point to an increase in slum dwellings in regions that faced larger tariff reductions. The magnitude of the slum coefficient is substantial: a municipality moving from the median to the 95th percentile of the tariff exposure distribution would experience an increase in slum households equivalent to 90% of the 1991 baseline mean (0.015). The precarious household coefficient, however, loses statistical significance once we include 1991 Census controls, suggesting that compositional differences across regions, such as pre-existing income levels and urban population shares account for part of the cross-regional variation in precarious housing conditions. While the slum estimate is striking, it should be interpreted with caution. Our panel spans a ten-year period, and although the tariff liberalization was concentrated between 1991 and 1995, the longer window leaves room for confounding factors to influence the magnitude of our estimates. That said, we argue that the positive sign of the slum coefficient is consistent with the labor market evidence in the literature documenting the persistent effects of the reform on local labor markets (Dix-Carneiro and Kovak 2017; Dix-Carneiro 2014). Taken together, the estimates in Table 3 support the hypothesis that the tariff reduction episode shifted the composition of urban areas toward greater housing informality, as reflected in the growth of slum settlements in more exposed regions.

7 Conclusion

In this paper, I present evidence that local labor market shocks induced by trade liberalization can alter the urbanization equilibrium within Brazilian regions. The findings point to the formation of slums in response to deteriorating local labor market conditions, suggesting that policies supporting workers through housing and urban interventions are particularly relevant in contexts of rapid city growth and mounting housing pressure. Moreover, the evidence presented here suggests that urbanization patterns are not the product of natural forces; they are shaped by economic forces, as documented throughout the paper, and are therefore amenable to policy intervention. Housing demand, especially in larger cities where informal settlements are most prevalent, can be addressed through deliberate policy design. The spatial equilibrium is not immutable: if society so chooses, it can be reshaped in ways that better protect vulnerable populations and expand access to adequate housing for those who need it most.

water with internal connection; adequate sanitation, defined as connection to the general sewage network or a septic tank; waste collection by a public cleaning service; electric lighting; and a household density below three residents per bedroom. This definition follows the IBGE criteria for the ONU-Habitat SDG Indicator 11.1.1 (IBGE 2024), but adopts a more restrictive threshold: we require all five conditions to be simultaneously unmet to reduce noise arising from the high baseline rates of infrastructure deprivation in poorer Brazilian regions.

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(a) Urban Area from Rio de Janeiro

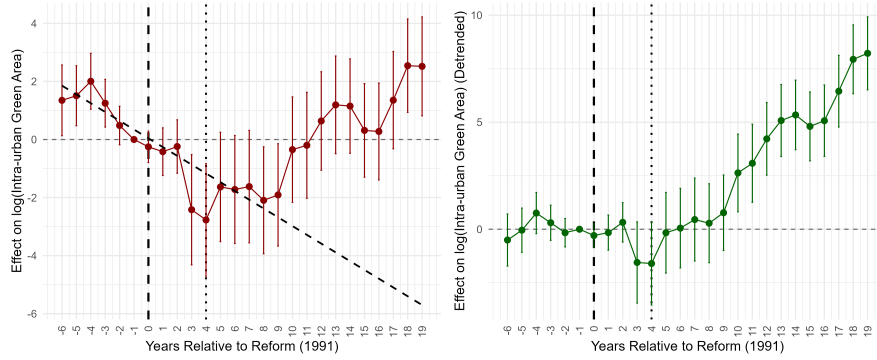


(b) Non Urban Area from Rio de Janeiro

Figure A.1: Urban and Non Urban Mask Layers from Rio de Janeiro

Source: MapBiomass Algorithm's Theoretical Basis Document (ATBD) (Collection 9, Version 1). **Note:** The Figure A.1a shows the urban mask built for MapBiomass based on pathways from OpenStreetMap database, representing all roads, streets, sidewalks, urban patches and conglomerates of specific categories (residential, service, path, and living street) . Similarly, the Figure A.1b shows The non-urban mask, which is the symmetrical difference of the final urban mask. The full description of the figure is available at <https://brasil.mapbiomas.org/download-dos-atbds-com-metodo-detalhado/>.

Figure A.2: Long-Term Effect of the Tariff Reform on Intra-Urban Vegetation in Urban Areas



Note: The figure presents the estimation of Equation 4 at the micro-region level. The dependent variable is the logarithm of intra-urban vegetation area located within urban boundaries. The specification includes time and micro-region fixed effects. The treatment variable is measured at the micro-region level, and the exposure period is centered on the 1990 tariff reform.

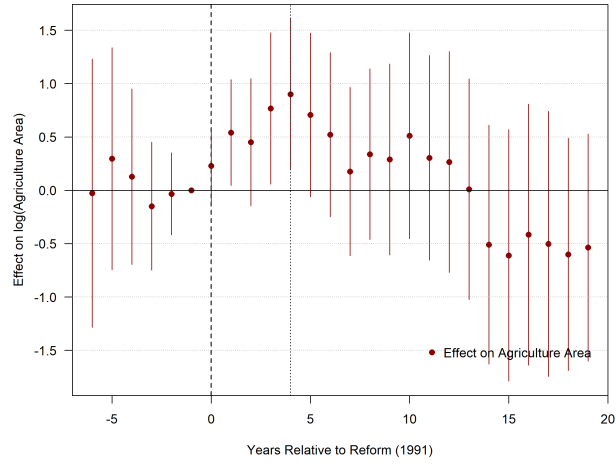


Figure A.3: Effect of RTR_m on $\log(Agriculture_{mt})$

Note: The figure shows the estimation of equation 4 at the micro-region level, with the dependent as the log of the agriculture area in urban census tracks. We include time and micro-region fixed effects. The treatment variable is at micro-region level.